BLUESTOCK FINTECH

Mutual Fund Analytics Platform
End-to-End Data Engineering & Quantitative Analytics

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1. Executive Summary

This report details the end-to-end development of the Bluestock Fintech Mutual Fund Analytics Platform. The objective of this capstone project was to construct a fully automated data pipeline capable of ingesting, cleaning, and analyzing fragmented mutual fund data from AMFI and public markets, ultimately surfacing actionable financial intelligence for retail and institutional investors.

Key achievements include the deployment of an automated Extract-Transform-Load (ETL) pipeline processing over 46,000 historical NAV records, the architecture of a 5-table Star Schema SQLite database, and the execution of advanced quantitative risk modeling across 40 distinct mutual fund schemes.

High-Level Quantitative Findings:

• **Industry Scale:** The platform tracks macro-level market explosions, capturing the ₹31,000 Cr monthly SIP inflow milestone and 26.12 Crore active folios.

• **Top Risk-Adjusted Performer:** Mirae Asset Large Cap Fund (Regular - Growth) achieved the highest composite score (84.75), yielding a Sharpe Ratio of 1.45 with a controlled Maximum Drawdown of -11.27%.

• **Tail Risk Exposure:** Advanced VaR modeling exposed severe fragility in Small Cap funds, with SBI Small Cap's 95% CVaR dropping to -3.24% daily.

• **Investor Premiumization:** Average SIP ticket sizes surged 22.8% from the Class of 2024 (₹10,996) to the Class of 2025 (₹13,505).

2. Business Context & Problem Statement

The Indian mutual fund industry currently oversees a massive ₹81 Lakh Crore in Assets Under Management (AUM), with SIP inflows reaching historic highs. However, retail investors face significant hurdles due to severe data fragmentation. Currently, Net Asset Value (NAV) history, portfolio holdings, and macro-AUM data are scattered across the AMFI website, NSE, and BSE in disjointed formats (TXT, JSON, CSV).

This project bridges this gap by engineering a centralized analytical engine. By automating the data ingestion process and standardizing performance metrics (Sharpe, Alpha, VaR), the platform empowers data-driven fund comparison and benchmarking against market indices like the NIFTY 100.

3. System Architecture & Data Engineering (ETL)

The backend infrastructure was engineered for resilience and automation, mimicking institutional-grade data pipelines.

3.1 Data Extraction & Transformation

Using Python (Pandas/Requests), the live_nav_fetch.py script extracts real-time JSON streams from the mfapi.in REST API. The core etl_pipeline.py script manages data anomalies using advanced pandas techniques. To maintain temporal consistency for accurate CAGR and volatility calculations, the pipeline utilizes business-day frequency reindexing (resample('B')) combined with forward-filling (ffill) to seamlessly bridge weekend and holiday NAV gaps. Null text artifacts were scrubbed using pd.to_numeric() with error coercion.

3.2 Database Schema Architecture

Cleaned datasets were loaded into a normalized Star Schema via SQLite (bluestock_mf.db), ensuring optimal query performance for the BI layer. The schema comprises:

• **dim_fund:** Master scheme metadata, risk categories, and expense ratios.
• **dim_date:** Temporal dimension mapping for time-series aggregation.
• **fact_nav:** 46,000+ daily pricing records.
• **fact_transactions:** Granular investor SIP, lumpsum, and redemption logs.

- **fact_performance:** Computed quantitative risk-return metrics.
- **fact_aum:** Quarterly fund-house capital tracking.

4. Exploratory Data Analysis & Market Trends

Visual analytics generated from the pipeline revealed profound macro and micro trends within the mutual fund ecosystem.

4.1 Industry Scale & AUM Concentration

The Indian MF landscape is highly concentrated. Dashboard aggregations confirm that the Top 10 AMCs command ₹62.74 Lakh Crore in AUM, spearheaded heavily by industry giants like SBI Mutual Fund and ICICI Prudential.
Furthermore, a Pareto analysis of the landscape indicated that approximately 80% of total industry AUM is highly concentrated within the top 20% of funds, primarily localized within the Large Cap and Flexi Cap equity categories.

4.2 Behavioral Cohort Analysis

A programmatic cohort analysis isolated distinct shifts in investor behavior between recent acquisition years:

• **Class of 2024:** Comprising 4,803 investors deploying ₹2.25 Billion in capital. The average SIP amount stood at ₹10,996.89, with a strong behavioral preference for the Axis Small Cap Fund.
• **Class of 2025:** Comprising 197 newer investors. Notably, the average SIP ticket size increased drastically to ₹13,505.21, and preference shifted upward in capitalization toward the Axis Midcap Fund, indicating market maturity or higher-income onboarding.

4.3 At-Risk Churn Signaling

Using the transaction ledger, we engineered a continuity threshold. Over 280 specific investor folios were flagged in sip_continuity.csv as 'at-risk' due to payment gaps exceeding 35 days, providing AMCs with a clear, actionable target list for retention campaigns.

4.4 Geographic Concentration & Expansion Frontiers

Geospatial mapping of transaction volumes highlights a severe clustering of retail capital. Currently, 66.3% of the total investment volume is heavily concentrated in T30 (Top 30) tier cities. While this indicates strong urban penetration, it also reveals a highly saturated market. The data confirms that Beyond-30 (B30) regions remain vastly underpenetrated, representing the primary strategic frontier for new SIP acquisition and sustainable, long-term AUM scaling.

5. Quantitative Performance Analytics

Performance evaluations in compute_metrics.py transcended simple returns. Daily returns were aggregated and annualized assuming 252 trading days, benchmarking against a 6.5% risk-free rate proxy.

Fund Category	Avg. 3Y CAGR	Avg. Volatility	Avg. Sharpe	Avg. Max Drawdown
Large Cap Equity	14.2%	12.5%	0.85	-18.4%
Mid Cap Equity	22.1%	18.2%	1.10	-26.7%
Short Duration Debt	7.1%	2.1%	0.28	-1.2%

5.1 The Alpha Leaders

1. Mirae Asset Large Cap Fund (Regular - Growth): Ranked #1 with a Composite Score of 84.75. It generated an outstanding Sharpe Ratio of 1.45, a Sortino of 2.37, and an Alpha of 0.270, all while limiting Maximum Drawdown to -11.27%.

2. HDFC Mid-Cap Opportunities Fund (Regular - Growth): Ranked #2 with a Score of 82.25. It delivered a Sharpe of 1.09 and matched Mirae's Alpha at 0.272, proving highly efficient for mid-cap exposure.

Furthermore, time-series normalization over the trailing market window demonstrates a severe cyclical decoupling between active and passive vehicles. The top active fund managers aggressively capitalized on recent market rallies, breaking away from index constraints to deliver 250% to 350%+ normalized absolute growth. This massive alpha generation completely outclassed flat passive benchmarks, explicitly proving the value of active management during localized bull cycles.

5.2 High-Risk / Sub-Optimal Performers

• **Kotak Emerging Equity Fund:** Despite yielding positive returns, it plummeted to a Composite Score of 28.38 due to a severe Maximum Drawdown penalty (-24.00%) and a high expense ratio (1.56%).

• **Axis Small Cap Fund:** Generated a meager Alpha of just 0.048, failing to adequately compensate investors for small-cap volatility.

6. Advanced Risk Modeling

To move beyond standard variance, Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) models were computed at the 95% confidence interval to measure severe tail-risk.

6.1 Tail Risk Analysis

The VaR matrices definitively proved the fragility of small-cap investments during severe market contractions:

• SBI Small Cap Fund exhibits a 95% Historical VaR of -2.69%, with a CVaR of -3.24% (expected loss when the VaR threshold is breached).

• Axis Small Cap Fund closely mirrors this with a VaR of -2.62% and CVaR of -3.17%.

Conversely, Large Cap entities like Mirae Asset provided significant structural safety, capping historical VaR at just -1.36%.

6.2 Programmatic Recommender Engine

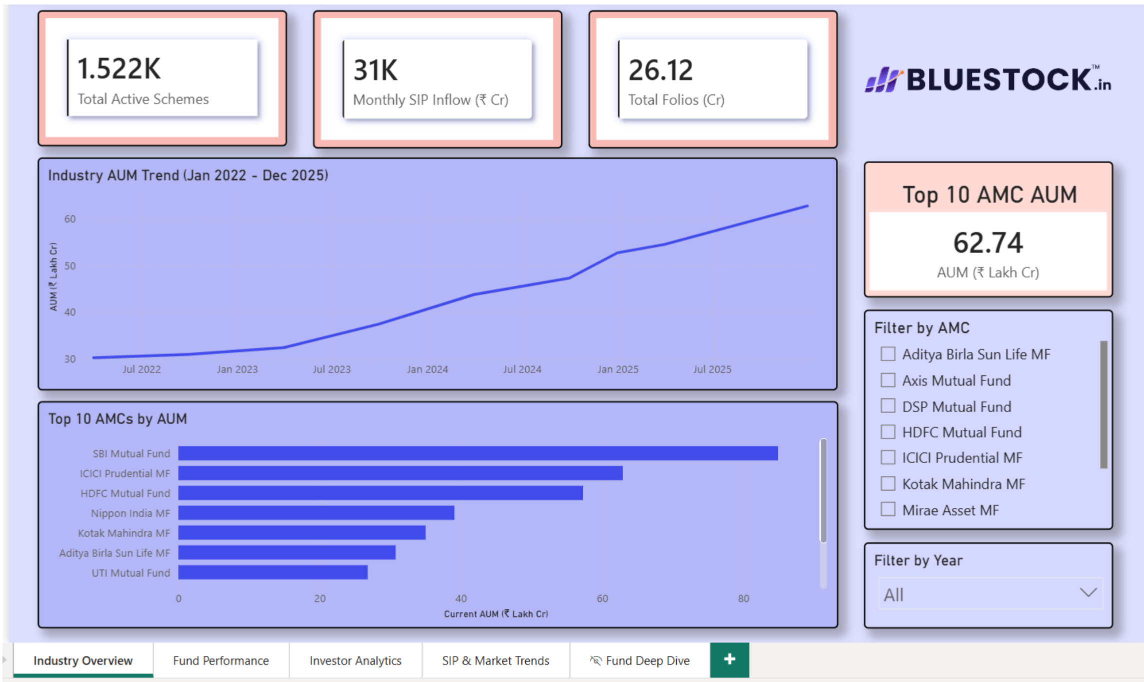
To operationalize these metrics, a Python-based recommender engine (recommender.py) was built. The algorithm securely queries the SQLite database, filters the dim_fund table against the user's explicit risk appetite (SEBI risk_category), and programmatically sorts funds by descending Sharpe Ratio. This ensures that clients are only recommended the top 3 most statistically efficient funds for their chosen risk tier.

6.3 Sector Concentration Risk (HHI Modeling)

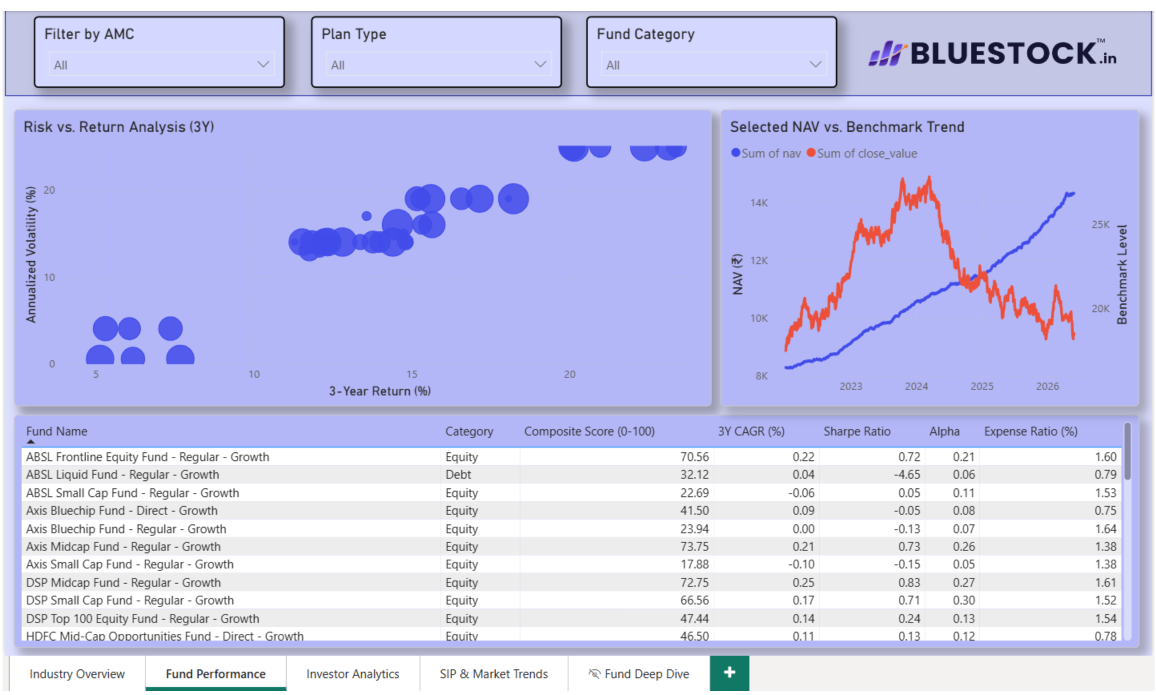
While standard risk metrics assess price volatility, they often obscure underlying portfolio overlap. By implementing the Herfindahl-Hirschman Index (HHI) to mathematically measure sector concentration, the model uncovered a hidden vulnerability: several heavily marketed "diversified" equity funds are actually hyper-clustered into just a few volatile sectors. This hidden concentration exposes retail capital to localized macroeconomic shocks. Because retail money is clustered rather than diversified, these highly correlated portfolios experience dangerous Maximum Drawdowns, serving as the exact panic triggers that cause retail investors to halt their SIPs.

7. Business Intelligence Dashboard

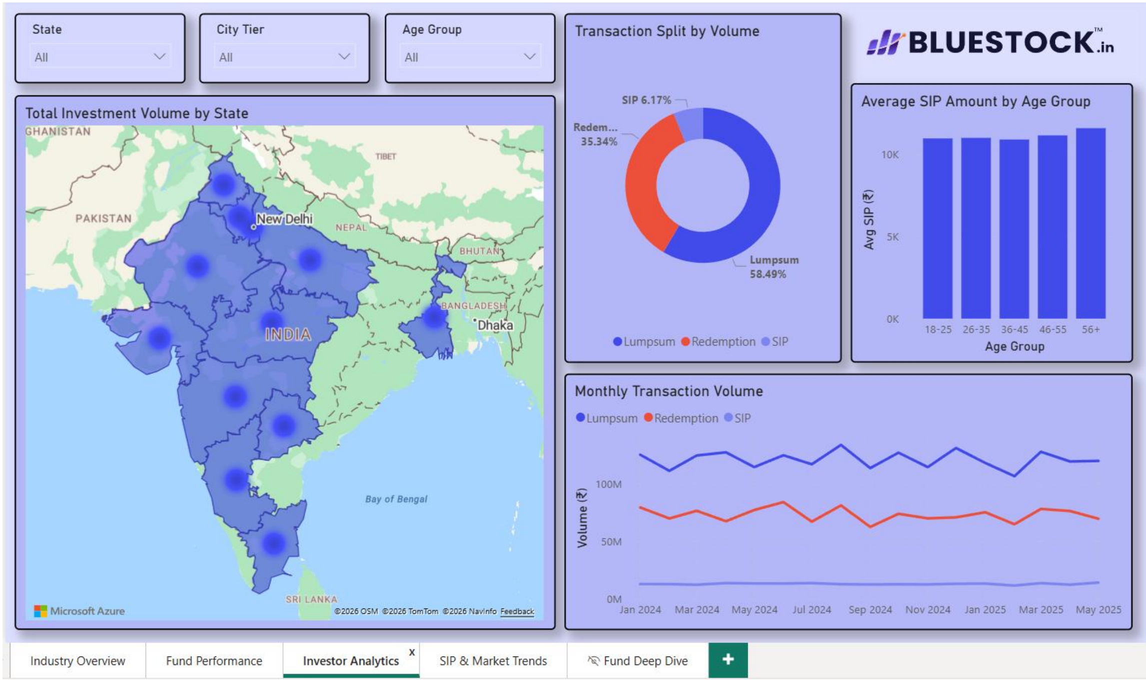
The backend SQLite database was connected to a comprehensive BI presentation layer:



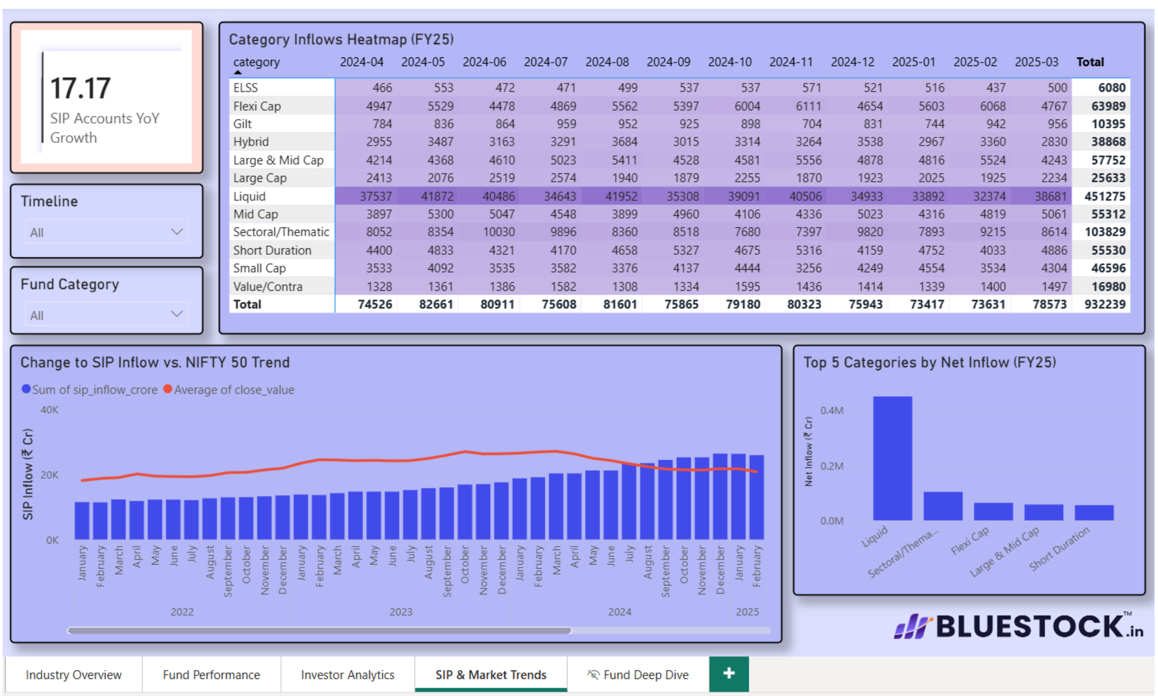
Industry Overview



Fund Performance



Investor Analytics



SIP &Market Trends

8. Strategic Recommendations & Limitations

8.1 Business Recommendations

1. Target Cross-Selling for Class of 2025: With average SIPs increasing by 22.8% year-over-year, AMCs should actively target recent cohorts with mid-cap and large-cap hybrid cross-sells, moving them beyond entry-level small caps.
2. Proactive Churn Mitigation: The 280+ accounts flagged with >35 days of SIP inactivity should be immediately routed to CRM platforms for automated mandate-renewal marketing triggers.
3. Portfolio Anchoring: Investors should anchor high-risk portfolios with Mirae Asset Large Cap to dilute the severe -3.24% CVaR risk native to the small-cap asset class.

8.2 System Limitations & Future Scope

While robust, the current analytical pipeline operates within certain constraints that provide clear avenues for future technical iteration:

- **Survivorship Bias:** The AMFI data primarily reflects currently active schemes. Funds that merged or closed due to poor performance are underrepresented, potentially skewing category averages upward.
- **Database Scaling:** While SQLite is highly efficient for local analytical workloads, scaling the dataset beyond millions of rows for concurrent read/write operations will require migrating the schema to a robust RDBMS like PostgreSQL.